

Axiom Studio

Physics you can actually touch.

Live: axiom-studio20.vercel.app · Built for DHS Hacks

PURPOSE

Every physics student knows the feeling: you memorize a formula, plug in numbers, get the "right" answer — and still have no idea what actually happened. Real labs are expensive and rare, so most students never get to **touch** the thing they're studying. Axiom Studio closes the gap between **knowing a formula** and **having intuition** by turning any browser into a real physics lab — no account, no setup, completely free.

HOW IT WORKS — THE LEARNING LOOP

1 · PREDICT

Commit to an answer before anything runs, so the "aha" lands when reality disagrees.

2 · EXPERIMENT

Adjust parameters with sliders and run a real simulation, not a canned animation.

3 · UNDERSTAND

Axion, an AI tutor, reads your exact numbers and prediction and explains why what happened, happened.

FOUR INTERACTIVE MODULES

Projectile Motion

Kinematics — find the launch angle that outranges them all (it peaks at 45°).

Pendulum

Harmonic motion — discover the period cares about length, not swing angle.

Collisions

Conservation laws — momentum always survives; kinetic energy may not.

Orbital Mechanics (3D)

Gravitation — the razor line between orbit, crash, and escape.

KEY FEATURES

- ▶ **Real physics engines** — simulations integrate the actual equations of motion every frame, using **4th-order Runge-Kutta (RK4)** integration so orbits stay stable and readouts stay trustworthy.
- ▶ **Axion AI tutor** — generates explanations tied to the learner's own run and prediction, not a generic textbook paragraph.
- ▶ **Prediction-first pedagogy** — locking in a guess before the sim runs is what builds genuine intuition.
- ▶ **Zero friction** — runs in any browser, free, with no account or install required.